

STEMI CENTER STANDARDS

Effective: February 15, 2025
Replaces: March 10, 2024

I. Purpose

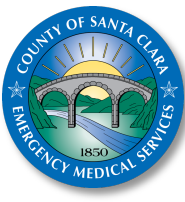
The purpose of this policy is to establish the procedures and standards for the Santa Clara County. Cardiac and ST- Elevation Myocardial Infarction (STEMI) Care System. This critical care system links pre-hospital and hospital care to ensure Cardiac and STEMI patients are taken to the closest, most appropriate hospital and receive timely, optimal treatment based on these standards.

II. Authority

California Health and Safety Code, Division 2.5, Sections 1997.220 & 1798
California Code of Regulations Title 22, Division 9, Chapter 6.2
California Code of Regulations Title 22, Division 9, Chapter 10

III. Definitions

- A. Cardiac Catheterization Laboratory or “cath lab” means the setting within the hospital where diagnostic and therapeutic procedures are performed on patients with cardiovascular disease.
- B. Percutaneous Coronary Intervention (PCI) means a procedure used to open or widen a narrow or blocked coronary artery to restore blood flow supplying the heart, usually done on an emergency basis for a STEMI patient.
- C. ST-Elevation Myocardial Infarction (STEMI) means a clinical syndrome defined by symptoms of myocardial infarction in association with ST-segment elevation on electrocardiogram (ECG).
- D. STEMI Patient means a patient with symptoms of myocardial infarction in association with ST-Segment Elevation in an ECG.
- E. STEMI Receiving Center (SRC) means a licensed general acute care facility that meets the minimum hospital STEMI care



requirements and is available 24-hours a day, to perform PCI.

- F. STEMI Referring Hospital (SRH) means a licensed general acute care facility that meets the minimum hospital STEMI care requirements to stabilize and transfer patients to a PCI-capable facility.

IV. The Santa Clara County EMS Agency (EMS Agency)

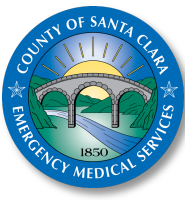
- A. The EMS Agency and EMS Medical Director are responsible for the oversight and management of the STEMI System including but not limited to planning, implementation and evaluation. Areas of responsibility include:

1. Designation and monitoring for compliance with standards.
2. Medical Control, clinical operations, training and management of the STEMI system in the pre-hospital care setting.
3. Developing pre-hospital screening, triage, and destination policies.
4. Coordinating STEMI system quality improvement.

V. STEMI Referral Hospital Standards

- A. Any acute care facility not designated by the EMS Agency as a STEMI Receiving Center shall meet the following requirements as an SRH:
1. Shall have written protocols and standardized procedures to identify STEMI patients and ensure optimal reperfusion strategies, including fibrinolytic therapy.
 2. Shall establish a transfer process which includes agreements for interfacility transfer with approved STEMI Receiving Centers and EMS ambulance providers to ensure rapid transport of STEMI patients to an SRC.
 3. Shall collaborate with STEMI receiving center(s) and the EMS Agency to review cases when requested and participate in the quality improvement processes.

VI. Designation

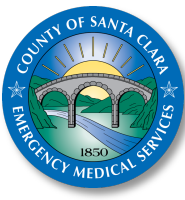


- A. The process for initial designation as a STEMI Center by Santa Clara County Emergency Medical Services are detailed in Policy #400.
- B. Continued designation as an EMS STEMI Receiving Center shall depend on:
 - 1. Current written agreement with the EMS Agency for designation as a STEMI Receiving Center.
 - 2. Currently serving in the EMS system as a Pre-hospital Receiving Center (PRC) Operate a cardiac catheterization laboratory licensed by the Department of Health Services and approved for emergency PCIs.
 - 3. Compliance with all State regulations and Santa Clara County EMS Agency policies and procedures detailed herein.

VII. STEMI Receiving Center Standards

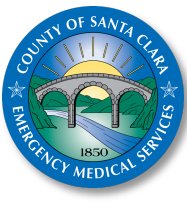
A designated SRC shall meet all the following standards:

- A. Hospital Facility and Services
 - 1. Agree to accept all EMS suspected STEMI patients according to applicable EMS policies/protocols.
 - 2. Have established protocols for triage, diagnosis, and cath lab activation and treatment of both EMS and self-transported STEMI alert patients.
 - 3. Maintain adequate number of competently trained medical staff, clinical supervisors, support staff and paraprofessionals to support cardiac services offered.
 - 4. Have a process in place for the treatment and triage of simultaneously arriving STEMI patients.
 - 5. Have the ability and process to receive ECGs wirelessly transmitted by pre-hospital personnel.
 - 6. Maintain uninterrupted STEMI capabilities which include advanced imaging, laboratory services, cardiac catheterization lab available for treatment of STEMI patients twenty-four (24)



hours per day, seven (7) days per week, three hundred and sixty-five (365) days per year.

- a. The hospital shall maintain a single call STEMI team and Cardiac Catheterization Team call rosters that may be requested by the EMS Agency for audit.
 7. Not exceed more than 100 hours of bypass per quarter.
 8. Have a protocol to be used in circumstances when PCI of a STEMI patient is not possible, or delay of the cath lab team to the patient exceeds 30 (thirty) minutes.
 9. Have mechanical circulatory support device (IABP, Impella, ECMO) immediately available for the cath lab and intensive care unit.
 10. Have available within two (2) hours of arrival, cardiac surgery services including operating room availability either directly or under written transfer plan and agreement.
- B. Staffing Requirements
1. STEMI Medical Director
Designate a board-certified physician in cardiology, cardiovascular surgery, interventional cardiology or another board with sufficient experience and expertise managing cardiovascular disease. They provide medical oversight of the STEMI program while also ensuring compliance with all applicable status, regulations and policies. They collaborate with the program manager for quality improvement initiatives, patient safety, medical staff training and education.
 2. STEMI Program Manager
Designate a licensed registered nurse experienced in the clinical care of STEMI patients and provide administrative support, coordination, and collaboration with the STEMI program medical director to ensure quality improvement, education and training of the STEMI and cath lab teams, ancillary staff and patients. They are responsible for compliance with State statute, regulations and County policy along with the program Medical Director.
 3. Clinical STEMI Team



Designate a multidisciplinary group of personnel to collectively provide care to patients experiencing myocardial infarction. Specially trained health care professionals that perform percutaneous coronary intervention. It may include, but is not limited to, an interventional cardiologist, midlevel practitioners, registered nurses, technicians, and other health care professionals. Cardiac catheterization team staff must have the required training and certification to work in a cath lab in California.

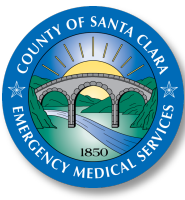
4. Registrar
Designate an individual to enter cases into the registry, pull reports and assist with quality improvement and performance indicator metrics.

C. STEMI System Participation

1. Participate in the pre-hospital performance improvement process and offer expertise to the EMS Medical Director in the development of pre-hospital care protocols relevant to STEMI patients.
2. Participate in the Cardiac Care System Quality Improvement Committee (CCSQIC) as per Policy #431 and other related committees when requested.
3. Participate in the pre-hospital quality improvement process through selected review of cardiac-related deaths, major complications, and transfers when requested by EMS Agency.
 - a. The STEMI Program Manager or designee from each designated facility will work with the EMS Agency Specialty Programs Nurse Coordinator to obtain the necessary chart materials for the medical review process and committee meetings.
4. For any STEMI Center receiving a STEMI patient as a result of ED Bypass or STEMI Bypass shall conduct a quality review on the patient and report any adverse finding to the EMS Agency.

D. Data Collection and Submission

1. The EMS Agency will be responsible for creating a *Santa Clara County STEMI Data Dictionary* to include the required local and state data elements and will be updated annually.



2. SRC Shall:

- a. Collect the minimum data elements identified in the *Santa Clara County STEMI Data Dictionary* shall be abstracted directly into a registry approved by the EMS Agency.
 - b. Submit the data to the EMS Agency on regularly assigned dates.
 - c. Ensure that data is validated and without error prior to submission.
3. STEMI data will be submitted to the California EMS Information System on a quarterly basis by the EMS Agency, as required by State regulations.

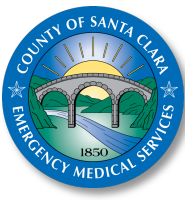
E. Performance Standards

A designated SRC shall meet all the following performance standards:

1. Have a process to review and revise facility protocols the identification and treatment of STEMI patients in the emergency department, cath lab, and intensive care unit at least every three (3) years.
2. Have a data-driven quality improvement program that monitors standardized clinical performance measures which align with American Heart Association/American College of Cardiology recommendations.
3. Additional performance measures as determined by the EMS Agency and/or the CCSQIC.

F. Education

1. Provide pre-hospital cardiac care continuing education opportunities at least once per year for EMS personnel. CE activities must be reported to the EMS Agency by the end of each calendar year.
2. Ensure staff physicians, nurses, cardiac cath lab staff and members of the STEMI clinical team receive annual training and competencies based upon the needs of the facility.



3. Have a patient population focused community outreach and education campaign the details heart attack warning signs and the importance of early utilization of the 9-1-1 system. Activities must be reported to the EMS agency by the end of each calendar year.

VIII. Reporting Requirements

- A. Shall notify the EMS Agency Duty Chief as soon as possible in the case of an unplanned event causing the cath lab or cardiac interventional to be unavailable as well as when the issue is resolved.
- B. Shall notify the EMS Agency through the submission of a System Variance Report at least 24 hours prior to any planned event that will result in the cath lab or cardiac interventional radiologist being unavailable. Shall notify the EMS Agency in writing of any failure to meet these EMS STEMI Center Standards within 48 hours. Shall notify the EMS Agency in writing of any failure to meet these EMS STEMI Center Standards within 48 hours.
- C. Shall notify the EMS Agency in writing of any failure to meet these EMS STEMI Center Standards within 48 hours. Shall notify the EMS Agency in writing of any failure to meet these EMS STEMI Center Standards within 48 hours.
- D. Changes to key SRC personnel shall be reported to the EMS Agency within 10 business days, including:
 1. STEMI Medical Director.
 2. STEMI Program Manager.

IX. References

- A. Writing Committee Members, et al. "2021 ACC/AHA/SCAI guideline for coronary artery revascularization: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines." Journal of the American College of Cardiology 79.2 (2022): e21-e129.